

RENANCE

PETROLEUM TRAINING INSTITUTE, EFFURUN

PET 101 — Past Questions & Materials

by Renance

Course: PET 101

Institution: Petroleum Training Institute, Effurun

Papers in this pack: 1

Levels: ND1

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Paper 1 of 1 — PET 101

Level ND1 • First Semester • Handout

PET 101 / MPE 107

INTRODUCTION TO PETROLEUM TECHNOLOGY

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Aim – To provide an overview of petroleum production technology.

At the end of the course the trainee should have a clear understanding of various processes involved in petroleum exploration, drilling, production and terminal operations.

General Objective:

- 1.0 Understand the term Petroleum, its origin and the properties of reservoir rocks.
- 2.0 Understand the term Petroleum exploration and its various methods.
- 3.0 Understand the term Drilling, oil and gas wells, drilling rigs and drilling methods.
- 4.0 Understand Drilling rig components and the various hole problems
- 5.0 Understand Well completion and subsurface operations activities
- 6.0 Understand Surface Gathering Systems, Transport and Storage of Crude oil and gas.

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Course Content

- Introduction
- Origin of petroleum
- Exploration methods
- Drilling operation
- Drilling Hole problems
- Well completion operations
- Flowstation operation
- Terminal operations

Recommended Texts

- 1) Fundamentals of Petroleum by Kate Van Dyke
- 2) Fundamentals of Drilling, Technology and Economics by John L. Kennedy
- 3) A Primer of Oilwell Drilling Fifth edition by Ron Baker
- 4) Drilling and Well Completion by Karl Catlin
- 5) Petroleum Drilling Series, University of Texas

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(WEEK 1 - 2)

CHAPTER ONE

Introduction

For any nation to be industrialized there must be sufficient energy to drive the economy vis a vis the industry. The early industrialized nations depended solely on coal energy to drive their engines. This later gave way to the use of steam as energy source. The fossil fuel was later discovered as a more efficient form of energy. The fossil fuel today has become so important that countries go to war to either claim ownership of where it can be found or protect its location within her territory.

Though alternative energy source is being sought since the fossil fuel is not renewable, but that remains a challenge to the scientist, since without efficient energy source most economies will be grounded.

This singular factor make the study of science of petroleum becomes more important both to the countries that have it and to those that seek alternative energy source.

ORIGIN OF OIL AND GAS

ORIGIN OF PETROLEUM

Petroleum simply means stone oil. The fluid mineral petroleum is an extra-ordinary unique substance. Its history dates back to early civilization. It is a naturally occurring mixture of complex hydrocarbon that exists as crude, natural gas, tar, and asphalt containing some impurities like sulphur, hydrogen sulphide and some mechanical impurities like sand.

Petroleum occurs in sedimentary basin especially where there is very thick sediment and there is completely no oxidation, under this condition the organic matter and sediments are converted to hydrocarbon.

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Inorganic Theory

Early theories about the origin of petroleum though it was abiogenic, i.e. formed from inorganic materials through chemical reactions. Although there has been more interest in the abiogenic origin with sophisticated theories, it does not seem to be correct for the following reasons:

- 1) None of the reactions have been known to produce the complex mixture present in crude oils. E.g. $\text{CaC} + 2\text{H}_2\text{O} \rightarrow \text{CH}_4 + \text{CaO}_2$
- 2) Petroleum has a relatively shallow habitat and therefore not probably formed as a deep seated magmatic product.
- 3) Its compounds are heat sensitive.
- 4) It occurs most frequently in sedimentary rocks.

The initial idea was based on the fact that methane gas, an important constituent of hydrocarbon is also highly associated with the atmosphere of outer planets of the solar system and their satellites.

Organic Theory

Modern theories now suggest that petroleum is of biogenic origin. That is must have occurred by the bacteria activity on the remains of animals and planets buried on heaps for many years. Evidence in support of this includes:

- (i) Crude oil is highly associated with beds containing the remains of marine organisms.
- (ii) Almost all living matter contains lipid fatty acid which could be acted upon by bacteria to produce paraffin hydrocarbons.
- (iii) The presents in crude oils of certain compounds which would have been destroyed in the presence of oxygen also suggest that petroleum must have been formed through the anaerobic activity of bacteria to the remains of living things.

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The biogenic origin can further be explained in this way. Elements of weather temperature change, wind and rain wears away the surface of the rocks and transport them to lower areas such as sea beds. The particles are usually deposited where the water transporting them becomes sufficiently calm. These depositional features form the marine sediments in which petroleum is formed. The river water transporting the sediments also has great quantities of microscopic plant and animal life. These with planktons living in sea water settle to the bottom of the sea at a place where other sediments continue to deposit on top. This sediment is later acted upon by anaerobic bacteria under heat and pressure to generate petroleum. Considering available facts, petroleum is more of biogenic origin as the process still takes place today.

MIGRATION PROCESS

After the oil has been formed, the next stage is migration. This is the process by which the hydrocarbon leave the fine grained rocks (source rock) in which they have been formed to enter the coarse grained rocks (reservoir rocks) in which they subsequently concentrate and segregate. The important reservoir characteristics essential for crude oil migration are porosity and permeability.

Porosity is a measure of the voids in a rock. It is expressed as a fraction of the total volume of the rock is percentage

i.e. porosity (%) = $\frac{\text{Total voids in a rock}}{\text{Bulk volume of rock}} \times 100$

Bulk volume of rock

In reservoir rocks, we talk about the effective porosity. This is the ratio of all the interconnected pores to the volume of the rock. It is also expressed in percentage. That means, for migration to take place, there must be effective porosity that is the pores of the rock must be interconnected.

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The permeability (K) of a reservoir rock i.e. its ability to transmit its fluid contents. A rock is permeable by reason of its effective porosity. This is because the pores

have to be interconnected before the fluid transmission.

PETROLEUM ACCUMULATION

Petroleum normally migrates to a reservoir rock which has the properties making it possible for it to concentrate, segregate and accumulate. It therefore means that after migration, we have accumulation. A major characteristic of the reservoir for accumulation of petroleum is a geological trap. The reservoir must be bounded by an impervious low permeability layer which makes the oil remain when it has migrated into the rock. The types of traps responsible for creation of barriers which necessitate petroleum accumulation are of different type but generally one, two or more traps may combine to form one where petroleum may later accumulate.

In summary, the necessary factors for petroleum to accumulate are:

- (1) A source rock effective porosity of a reservoir rock
- (2) Permeability for migration to take place and
- (3) A geologic barrier or trap for it to accumulate.

GEOLOGICAL TRAPS

The subsurface rocks in which hydrocarbons are trapped are referred to as reservoir rocks. These rocks have characteristics that make them unique in nature, acting as physical barriers to further migration of oil. A suitable reservoir rock is a necessity for hydrocarbon accumulation. A geologic trap which is usually an important

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feature of a reservoir rock could be of different types depending on the mode of formation. Those due to the deformation of earth (structural) e.g. fault traps, anticline trap, etc and that due to permeability difference (Stratigraphic). Other types of traps may be generated due to a combination of the above types and they are known as Combination Traps. Generally, no matter how the traps are generated, reservoir rocks are porous and permeable with impervious boundary or cap rock. Geologic traps are mainly due to earth movements. They are called traps because they act as physical barriers to further oil migration.

STRUCTURAL TRAPS

These are generally due to the deformation in the structure of the earth. It could be due to earth movements or faults. Almost all forms of traps have elements of Structural deformity. Structural traps are of different types but only few common examples will be considered here:

- (i) Anticlines:- These take the form of upfolds of rocks which are oval in shape in horizontal plane. They are the commonest among other types of structural traps in which crude oil accumulation has been found.

Gas

Oil

Water

Water

- (a) Fault Trap
- (b) Anticline Trap

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Fig. 1.1: Fault Trap and Anticline Trap

(ii) Fault Traps: - When strata are relatively displaced due to rock fracture, a fault results. Faults are due to shearing forces related to earth's movement and vary in size from very small local dislocations to major breaks more than hundred miles long. In general, faults produce oil accumulation when it acts in combination with other structural and stratigraphic trap.

STRATIGRAPHIC TRAPS

These are traps formed by changes in lithology, generally a disappearance of the containing bed or porosity zone. It includes traps bounded on one or more sides by low permeability zones. Almost all forms of stratigraphic traps also have structural factors in conjunction. We have basically primary and secondary stratigraphic traps. Primary Stratigraphic traps result from the formation of a zone of high permeability during the deposition of the rocks. Secondary Stratigraphic traps or secondary (unconformity) traps result due to the presence of an unconformity is a break in the sequence of local geological deposition. Traps may be formed below and above the plane of unconformity. See fig. 1.3.

Oil
Oil
Oil

Fig. 1.2:- Stratigraphic Trap

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(i) Combination Trap: - When two or more types of trap combine together to form another trap for hydrocarbon accumulation, it is called a combination trap e.g. a faulted anticline fig. 1.4.

Oil

Fig. 1.3:-Faulted anticline (Combination Trap)

(ii) Barren Trap: - A trap in a reservoir that contains no oil is called a barren trap. A trap is called an aquifer if it contains only water.

WEEK 3 - 5

CHAPTER TWO

EXPLORATION METHODS

Once the factors for petroleum accumulation have been known and the properties for oil bearing rocks understood, it is now time to find out how these rocks can be located. The methods for locating possible oil accumulation structures or minerals

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are known as exploration methods while those specifically used for petroleum exploration are known as petroleum exploration methods.

GEOPHYSICAL SURVEYS

Geophysics as a science applies the principles of physics to the study of the earth. Geophysical surveys involve taking measurements at or near the earth's surface that is influenced by distribution of the physical properties which when analyzed can give insight into the variation of the properties of the earth laterally and vertically. It is infact, the methods in which the properties of oil bearing rocks can be detected from the surface. Geophysical methods are capable of detecting and delineating local features of potential interest if properly applied. This will help a great deal in cutting down on costs avoiding unnecessary drilling. The benefits of geophysical surveys are so enormous that one form or the other is required as a pre-requisite before embarking on any exploratory drilling. Petroleum geologists are most interested in the earth's magnetism, gravity and especially seismic vibrations or vibrations of the earth. Sensitive instruments can measure variations in one of these physical qualities that may be related to conditions under the surface. These conditions, in turn may point to probable oil-or gas- bearing formation.

Hence, geophysical survey involves taking measurements at or near the earth's surface that are influenced by distribution of the physical properties. There are two broad divisions of geophysical survey viz the natural field methods (gravitational, magnetic, electrical and electromagnetic) utilizing the natural fields of the earth and the artificial source methods which involves the generation of local electrical or electromagnetic fields. Geophysical methods are many but four major types will be considered here. They are:-

- The seismic method which utilizes travel times of reflected/refracted seismic waves.

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- The gravity method: using variations in the strength of the earth's gravitational field.
- The magnetic methods: using variations in the strength of the magnetic field
- The electrical (resisting) method utilizing earth resistivity.

Geophysical methods are usually used in combination as some are usually used for reconnaissance - to select area for further studies. Thus it is usually in stages. For example, initial gravity survey may reveal large deposit of sedimentary rocks which can then be studied by the more detailed seismic survey.

Types of Geophysical Surveys

Magnetic Survey

Geologists searching for petroleum have made good use of the fact that the earth has

a strong magnetic field. Applying the principles that like rocks have similar magnetic fields enables the geologist to compare slight differences in the magnetism generated by the minerals in the rocks. These differences suggest the locations of different formation.

For example, igneous rocks, frequently found as the basement rock under sedimentary layers often contains minerals that are magnetic. Such rocks seldom contains hydrocarbons. But this rock type may sometimes intrude into the overlying sedimentary rock, creating arches and folds that may serve as hydrocarbon traps. Magnetic survey can be performed on land, marine environment, and air (aeromagnetic survey). The instrument used for magnetic survey is the magnetometer. Magnetometer detects slight variation in the earth magnetic field. These variations may show how deep magnetized rocks are buried. A magnetometer can be towed behind a ship or an airplane to cover large areas. It transmits data to a device on board which records the information onto paper or magnetic tape.

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Electrical Method

This method is based on the fact that rocks of different composition have different electrical properties. Two methods are generally employed in electrical method; these are natural electrical field of the earth and artificially induced currents. The most commonly employed electrical method is the artificially induced current or direct current method which measures the rock resistivity. In this method artificially generated electric currents are introduced into the ground and the resulting potential difference is measured at the surface utilizing four electrodes altogether. This is Wenner electrical configuration.

V

q q q

E1 P1 P2 E2

Fig. 2.1 Wenner Electrical Configuration

Most rock forming minerals are insulators and only conduct electricity through the ions in pore spaces by water molecules. Therefore porosity is a major control of the resistivity of the rock. Resistivity generally increases as the rock porosity decreases. The amount of detail result obtained depends on the distance between survey sites. Closely spaced sites results in a detailed survey, which is helpful in deciding where to drill new wells near as area of proven reservoir.

Gravity survey

Geophysicists also make use of slight variations in the earth's gravitational field caused by the varying weight of rocks. Some rocks are denser than others. Very dense rocks close to the surface exerts a gravitational force more powerful than that of a layer of very light rocks. The basis of gravity survey is Newton's law of

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gravitational which states that, the force of attraction between two masses m_1 and m_2 whose dimensions are small with respect to the distance r between them is given by:

$$F = G \frac{m_1 m_2}{r^2}$$

$$F =$$

$$r^2$$

G = gravitational constant.

Gravitational field is measured by very sensitive instrument called gravimeter. The unit of gravity is micrometer per seconds (μms^{-2}). Many types of corrections are done for different effects such as terrain, elevation e.t.c. before the interpretation of gravity survey.

The major setback with the gravimeter is that it responds only to vertical component of gravitational attraction.

THE SEISMIC SURVEY

Seismic survey is usually the last exploration step before an oil company actually drills a prospective site. Unlike gravity, magnetic and electromagnetic surveys that provide general information, seismic surveys give the explorationist more precise details on the formations beneath the surface.

The seismic survey is based on the fact that the earth crust has many layers with different thicknesses and densities. When energy from the surface, such as explosion strikes the layers, part of it travels through the layers and part of it is reflected back to the surface.

The velocities of seismic pulses or waves are determined by the elastic module and densities of the materials through which they pass. There are two types of seismic waves,

- (i) Body wave and
- (ii) Surface wave

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The body waves comprise of the compressional waves (the longitudinal, primary or P waves) and the shear waves (the transverse, secondary or S waves) while the surface waves can propagate along the boundary of the solid. The body waves propagate through the body of an elastic solid.

Seismic survey starts with small artificially produced 'earthquake' or explosions. They are received by seismic detectors which are capable of converting the mechanical input into electrical output. Sensors called geophones pick up the reflected seismic waves and send them through cables to a recorder. The recorder, a seismograph, amplifies and record their characteristics to produce a seismogram. The seismogram generates a seismic section, which is a two - dimensional slice from the surface of the earth downward.

The information from a seismic survey indicates the types of rock, their relative depth and whether a trap is present. The seismic survey is mainly dependent on the

differences that exist in the velocities of seismic wave through different rocks. The interpretation of seismic data is highly technical and requires the use of highly skilled personnel and the analytical application of mathematics, physics and geology. The type of seismic section where by subsurface features can be viewed in two dimensions is called 2-D seismic section. The seismic section where the subsurface can be viewed in three dimensions is called 3-D seismic section. The advantage of 3D seismic is that, with the help of computers, an explorationist can slice the cube in any direction north – south, east – west, horizontally or on any other plane. Focusing on an area in this way provides much more reliable information about the geologic structures it contains. Definitely, the three dimensional is more detailed than two dimensional seismic survey. 4D seismic survey is also available, this refers to repeated 3D survey (through time – the fourth dimension) to monitor changes in the formation, primarily changes in the fluid levels.

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AMBIGUITY IN GEOPHYSICAL SURVEYS

The internal structure and properties of the earth are not clearly known and this is the basic problem of geophysical interpretation. Thus, it is not possible to predict any geophysical method taken at the surface uniquely. For example, if it is required to determine the depth of a buried geological interface using seismic survey, the travel time of a pulse reflected from the surface is used. The velocity of wave in this medium must be known but this is not so. Therefore, assumption of any velocity to work out the result represents one of the possible solutions. This problem is usually solved to some extent by taking additional measurements but ambiguity in interpretation remains. The ambiguity problem in geophysical interpretation actually arises because many geological configurations could reproduce almost some observed measurements. In spite of the above limitations, geophysical surveys remain invaluable in petroleum exploration.

WEEK 6 - 9

CHAPTER THREE

DRILLING AND DRILLING PROCESSES

Once the probability of finding oil in a particular place is high after all initial exploration methods, it is now time to carry out operations to determine the actual occurrence of the hydrocarbon, which is exploratory drilling.

Drilling – Drilling can be defined as the application of science and technology to make a usable hole to the operator safely and economically

EXPLORATORY AND APPRAISAL DRILLING

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Once oil has been struck, there is the necessity to find out whether it is in a

commercial quantity or not.

The well drilled to find out the possibility of finding oil in a delineated area is called an exploratory well. This well is drilled for the purpose of ascertaining or confirmation of oil accumulation.

However, wells are drilled to determine the extent of the reservoir, so as to know volume of the oil reserve; such wells are referred to as appraisal well

In fact the first exploratory well is actually called a wild cat. As far as oil and gas deposits remains elusive, all wells drilled are exploratory wells.

DEVELOPMENT DRILLING

If appraisal and exploratory wells prove successful, that is, oil is struck in a commercial quantity, development drilling is embarked upon. This is the drilling for oil and gas and its subsequent production. A programme for the development of the oil field is planned during this period. The facilities and pattern of development are also planned. The number of wells and location based on the drive mechanism is planned for optimum capital utilization. The estimated income/expenditure and spacing for the wells are well put into consideration.

DEVIATED DRILLING (DIRECTIONAL DRILLING)

Directional Well - These are purposely drilled to deviate at a predetermined angle to a given target. This shows that a hole can be drilled vertically and also directionally and of course horizontally.

So we can say directional drilling is the science of directing a wellbore along a predetermined trajectory to intersect a designated sub - surface target. Horizontal Well drilling is the science of directing a wellbore along a predetermine trajectory usually at an angle of 90o to the vertical to intersect a designated sub- surface target

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Directional drilling is mainly applied when the oil bearing strata is under dwellings, industrial buildings and constructions or areas that cannot be drilled vertically due to some factors like mountains, underneath the floor of a sea, lake, river channel, floating barges, etc.

TYPES OF RIGS

A rotary rig may be thought of as a factory or manufacturing plant. It is designed to produce only one product, which is the hole.

It is always desired to drill or make hole in different geographic and climatic environment. The types of rigs employed for the hole making will be different from one environment to the other. This is to provide means of adapting the rig to a particular environment. However, the basic drilling operation remains essentially the same regardless of the types of rig used or the environment where the rig is used.

The location may be on land, Inland (swamp) and sea (offshore). The primary difference between lands, water-drilling installations occurs in the supporting structure for the rig and its auxiliary equipment. Other rig components are similar to

those used in land drilling with added emphasis on operational case safety precautions and standby equipment.

LAND LOCATION (ON SHORE)

For drilling on land (on shore) a land rig is used which consist of derrick or mast, substructure, mud pit, drillers cabin, prime movers, hoisting equipment, rotating equipment, circulating equipment and crew residential cabin.

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WATER DRILLING EQUIPMENT

Inland water equipment – Inland water drilling includes swamp, dredged marsh, bay and lagoon locations where water is relatively shallow and protected from appreciable wave action.

Hence, submersible drilling barge is used (Inland Barge). The submersible barge is a platform capable of supporting the drilling equipment, which may be towed from location to location with a minimum of lost time. The barge is then sunk, levelness being attained by controlling the amount of water in each barge compartment. After the well is completed, the procedure is reversed, and then the entire assembly is towed to the next location.

Inland barges are designed for drilling in 10 – 20 ft of water. Drilling in deeper water with submersible barge is accomplished by construction of a shell gravel fill for the barge to rest on.

OFFSHORE LOCATION

A self – contained platform is used, if the water is not too deep. The platform is sufficiently large to house the entire rig, crew quarters and supplies. An example of this is Jack – up rig, which may be used up to 350 ft of water. They are very useful in development drilling and poorly adapted to exploratory work.

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Fig: 3.1 Jack up Rig

Semi submersible Rig

These types of rig are often refers to as semi for short. They are floating barge. They use anchors on the sea floor to keep the rig stationary. They can be used in water up to 7000ft (2100m). Water is allowed to enter the barge so as to lower the rig to the sea level for drilling. Also water is removed from the barge after drilling in order to tow it to a new drilling site by a tugboat.

Fig.: 3.2 Semi submersible Rig

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Drilling Ship

This is a large self contained ship, which contains substructures, drawworks, rotary table, engines, mud pits, dry mud, chemicals etc, pipe racks, cementing equipment, fuel and water and crew quarters.

The ship can operate in any weather short of hurricane. It can be used for a deep offshore drilling and in remote areas. It uses dynamic positioning to keep them on station (it uses computer controlled motors or thruster and sensors)

Fig. 3.3 : Drill ship

DRILLING METHODS

Drilling methods can broadly be divided into two viz: percussion and rotary drilling. In percussion, drilling is achieved by the impact of a free falling body on the bit while in rotation drilling; it is achieved by the rotation of the bit.

PERCUSSION DRILLING

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In this method, drilling is accomplished by the pounding action of a steel bit, which is alternately raised by a steel cable and allowed to fall, delivering sharp successive blows to the bottom of the hole. The original percussion drilling apparatus consisted of a spring pole anchored into the ground at an angle, with the bit suspended from the free end by a rope. The Chinese employed a number of men who alternately jumped on and off the spring pole beam from a ramp.

As more and deeper wells began to be drilled, efforts were made to improve the drilling equipment. Steam engines began to be used, walking beams replaced the spring pole steel cables replaced manila ropes and other improvement followed. The cable tool rig is composed of drill string, rig lines, prime movers, Bailer and sand pump. The cable tool drill string is made up of the bit, dill stem, jars and a rope socket enabling their attachment to the drilling line or cable.

Other surface equipments used include prime movers, Bailers and sand pumps, Derrick. Percussion drilling depth is limited and highly un-economical and is therefore not widely in use nowadays.

ROTARY DRILLING

In the Rotary method, the hole is drilled by a rotating bit to which a downward force is applied. The bit is fastened to and rotated by a drill string, composed of high quality drill pipe and drill collars, with new sections of joints being added as drilling progresses. The cuttings are lifted from the hole by the drilling fluid which is continuously circulated down the inside of the drill string through water courses or nozzles in the bit and upward in the annular space between the drill pipe and the bore hole.

The parts of a drilling rig may be broadly classified into five, namely:

- (a) The rotating system
- (b) The hoisting system

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- (c) The circulating system
- (d) Blow-out prevention system and
- (e) The power system.

MAJOR COMPONENT OF DRILLING RIG

Power system (prime mover) power system

Drawworks

Block and Hoisting

Drilling line

Swivel

Kelly/Top Drive System

Rotary Table Rotary system

Drill pipe

Drill collar

Bits

Drilling fluid

Mud pumps Circulating system

Mud tanks

The mud treatment (desilter, dessander, degasser)

Blowout Preventer

Accumulator

Choke Manifold Well Control Equipment

Diverter

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HOISTING SYSTEM

The major parts of the hoisting system consist of the drawworks, travelling block, crown block and drilling line. When a drill bit is worn out, the drill string is tripped out and the pipe break-out till the last pipe is removed and the bit is replaced. As the hole deepens, the drill string is break out at the Kelly, a new pipe connected and drilling resumes or a new pipe is connected to the TDS which has a pipe grabber from where new pipe is added.

Wall caving is reduced greatly in rotary drilling due to mud cake effect and hydrostatic pressure of drill fluid.

The process of drilling can be summarized as involving the following operations running the drill string in the well, operating the drill bit on the bottom hole, making connections as well is deepened and pulling the drill string out of the well to change the drill bit.

DRAWWORKS

The drawworks, sometimes called the hoist, is a big heavy piece of machinery that consists of a revolving drum around which the wire rope, called the drilling line is spooled or wrapped.

FUNCTIONS OF DRAWWORKS

1. To lift pipe out of the hole
2. To lower pipe back into the hole

3. To make pipe by spinning cat head
4. To break out pipe by spinning breakout cat head

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Wire rope is reeled or spooled, on a drum in the hoist. When the draw works is engaged, it turns and depending on the direction that it turns, either reels in the drilling line to raise the traveling block or lets out the line to lower it. Since the drill stem is attached to the block, the stem is thus raised or lowered.

The hoist contains brake systems which enable the driller to easily control a load of thousands of pound of drill pipe or casing. There may be at least 2 types of brake system, Mechanical and hydrodynamic brakes.

Fig. 3.4 Drawwork

BLOCKS AND DRILLING LINE:

The drilling line supports the entire drilling assembly by means of pulley – style devices and grasping implements, which are collectively called the overhead tools. Included among these are the crown block, traveling block, the hook, drilling line and the elevators.

Drilling Line

Drilling lines are made of steel wires that are made into strands and the strands into rope in action similar to braiding. The outer strands are laid spirally around a central core. The wire rope generally ranges from 1 1/8 to 1 1/2 inches

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A wire rope is itself a machine because it requires lubrication. Reeving is the treading of the line from draw works drum to the dead line tie-down passing alternately it over a sheave of the crown block and a sheave of the traveling block.

Crown Block

The crown block is an assembly of sheaves mounted on beams at the top of the derrick. Most crown block has 4 to 7 (seven) sheaves which may be as large as 5 – feet in diameter mounted in line on a common centre pin.

Fig. :3.5 Schematic diagram of a crown block

Traveling Block

The traveling block moves between an up and a down position on the line. It carries the hook that support drill string when drilling and the elevator bail that holds the elevator when tripping in or out of the hole.

ROTATING SYSTEM

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The rotating equipment from top to bottom consists of the swivel, the Kelly, the rotary table, the drill stem and the bit for a Kelly rig. It also consists of Top Drive System, the drill stem and the bit for a top drive rig.

SWIVEL

Swivel in a device joining two parts so that one or both can pivot freely. It is attached to the traveling block by a large bail.

FUNCTION OF SWIVEL

1. It supports the weight of the drill stem
2. It allows the drill stem to rotate
3. It provides a pressure – Tight seal and passage way for the drilling mud to be pumped down the inside of the drill stem

KELLY

The Kelly is a square, hexagonal or triangular piece of pipe, usually about 40 ft long, which serves the purpose of transforming the rotating motion of the rotating table to the drill string.

FUNCTION OF KELLY

1. It serves as a passageway for drilling fluid on its way into the drill pipe.
2. It transmits the rotating movement to the drill pipe and bit.
3. It may serve as well control equipment through upper and lower Kelly cock.

The Kelly fits into a corresponding square or hexagonal opening in a device called a Kelly bushing or drive bushing. The Kelly bushing fits into a part of the rotary table called the master bushing or rotates and as the Kelly rotates, the drill string and bit rotate.

Top Drive System. (TDS)

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TDS is sometimes called power swivel, because it connects to top of the drill string providing rotational torque to the string while simultaneously supplying drilling fluids and supporting the string weight.

The top drive is suspended from the hook connected to the travelling block and moves on a rail. It has its own heavy-duty motor. The motor turns a threaded drive shaft with the aid of diesel-hydraulic power unit (HPU) that connects directly to the top of the drill stem to turn it. This completely replaces the Kelly and the rotary table in initiating rotation in a Kelly rig. Thus in a TDS rotational motion is initiated from the top instead of rotary table.

THE ROTARY TABLE

The function of rotary table is to:

1. To turn and rotate the Kelly (or any equipment attached to it)
2. To support the weight of the drill string or the casing during make up or break out

When drilling is going on, the rotating turns to the right or clockwise.

Fig.:3.6 Rotary Table showing Kelly bushing and master bushing in place.

DRILL STRING

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The drill string is made up of the drill pipe and special heavy - walled pipe called drill collars. Each length of drill pipes is about 30 ft long and is called a joint of pipe. Drill pipe is attached to Kelly at the top and to the drill collar at the bottom.

FUNCTION

1. Permit rotation
2. Allow fluid passage to the bit.
3. Provide necessary length of the well drilled

Each end of each joint is threaded. The end of the joint with interior threads is known as the box, and the end of the joint into the exterior threads is called a pin. When pipe is made up, the pin is stabbed into the box and the connection is tightened.

DRILL COLLAR

These are heavy walled pipe of small internal diameter and large external diameter steel tubes, through which mud can be pumped. Drill collars are heavier than drill pipes and are used on the bottom part of the string to put weight on the bit. This weight presses down on the bit to make it drill. Drill collars are about 40ft long.

FUNCTION OF DRILL COLLAR

- To transfer rotary action to the bit
- To allow passage for drilling mud
- To put weight on the bit
- To provide stabilization for the bottom hole assembly

BITS

The drilling bit is attached to the drill collar through the bit sub. This is the part of the rotary that does the actual hole making.

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There are 3 general types of bits

1. The Drag bit
2. The roller cone or rock bits
3. The Diamond bits.

THE DRAG e.g. FISH TAIL.

It is attached to the lower end of the drill collar, It drills by shoveling action against the formation. It is used mainly to drill a soft formation, especially for a shallow formation drilling.

ROLLER CONE BIT

This is the widely used bit in the industry. It has cone - shaped steel devices called cones that are free to turn as the bit rotates. Most bits have 3-cones, but some have 2 and some have 4 cones

There are two types of Roller cone bits

- (a) The steel tooth bit and (b) Tungsten - carbide inserted bit.

Fig. 3.7 Roller cone bit (inserted bit)

DIAMOND BITS

They do not have cones or rotating part. Several diamonds are embedded into the bottom and sides of the bit. Since diamonds are so hard, diamond bits are specially suited for drilling hard rock formations.

CIRCULATING SYSTEM

For the rotary drilling system to function, fluid must be circulated downward through the drill stem, around the bit and upward in the annular space between the drill stem

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and the wall of the hole or the casing. At the surface, the returning fluid (mud) is diverted through a series of tanks or pits which afford a sufficient quiescent period to allow hot mud to cool down and any necessary treatment. In the last of these pits the mud is picked up by the pump suction and the cycle is repeated. The upper part of the swivel is connected to a flexible hose (rubber) through which the mud is pumped. The mud passes through the drilling string and comes out through the nozzles of the drill bit. It then picks rock cuttings and returns to the surface through the annulus of the drill string and casing. It settles in the mud pits after proper cleansing by special equipment and it subsequently pumped into the well and the whole process continues.

For the rotary drilling system to function, fluid must be circulated downward through the drill stem, around the bit and upward in the annular space between the drill stem and the wall of the hole or the casing.

The fluid is pumped from the mud tank, to the surface lines to the stand pipe, rotary hole, goose neck, swivel, Kelly, (top drive system) drill pipe, drill collar, then to the bit. The fluid comes out of the bit nozzles up the annulus of drill collar and annulus of drill pipe, up to the mud return line and through the shale shaker degasser, desander, desilter and finally back to the mud tank. A mixing hopper is placed between the mud tank and the pump, where additives are added to the mud for conditioning.

THE DRILLING MUD (FLUID)

The term “drilling fluid” includes air, gas, water and mud. Term “mud” refers to a suspension of solids in water or oil or of solids and droplets of one of these liquids (emulsion) dispersed in the other.

The drilling mud consists of 3 primary parts.

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- (1) The liquid phase (oil for oil based or synthetic mud and water for water based mud)
- (2) The colloidal fraction, is the (bentonite clay) reactive portion
- (3) The inert fraction, which consists of sand, weight material and other inert solids (barite).

Oil – emulsion mud is simply water based mud is which minute drops of oil are suspended. The major part of mud control is directed to the colloidal fraction; for

this is the part of the mud enriched by the addition of bentonite and other clays and improved by chemical treatment, or is damaged by contaminant.

MAKE - UP WATER

The water used for make - up and maintenance of water base mud is classified as fresh or salt water. If water contains large amount of calcium and magnesium ions it is termed hard water. Bentonite and other clays disperse more richly in soft fresh water than in hard fresh water or salt water.

THE COLLOIDAL FRACTION.

Wyoming bentonite is predominantly montmorillonite a clay mineral which disperses as very small particles in water, hydrates and greatly increases viscosity and reduces filtration.

YIELD OF CLAY.

The yield of clay is defined as the number of barrels of fifteen centipoises 15 (cp) mud that can be obtained from one ton (2000lb) of material.

FUNCTIONS OF DRILLING FLUID.

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1 Transporting cuttings to the surface. Liquid, air or gas in circulation moves rock chips, sands or shale particles out of a well as it moves up the annulus. The mud must be viscous enough to take the cuttings to the surface and thin enough to prevent chip hold down effect and excessive pump pressure. The solids in mud are separated at the surface by screening.

2 Cleaning the bottom of the hole.

A bit must have a clean surface on which to work when making holes, whether it is crushing or shearing the formation. If chips or cuttings are not swept away as they are formed, the bit bogs down and eventually the drill stem cannot be turned. For the bit to regrind the chips already broken off from the bottom of the hole is wasted effort, reducing available power for holes making. High velocity stream of flood for the jet nozzle, blast the bottom of the hole, creates turbulence that moves the chips from the face of the formation as fast as they are formed. 50-60% of pump power reach the nozzle.

3 Cooling the bit and lubricating the drill string

The combination of heavy bit weight, which may be in excess of 60,000 Lb and a rotary speed of about (50 -100 rpm) create heat due to friction in the bit bearings and abrasion of the formation against the teeth or blades. Unless the bit is well cooled, overheats and quickly wear out.

Fluid circulated around the parts of the bit removes the heats. Oily substance in the drilling fluid can reduce friction in the bit bearings and acts as lubricant between the drill steam and the walls of the hole. Oil-emulsion mud and oil mud are especially helpful in this way.

4 Supporting the walls of the wellbore:

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A drilling fluid with proper characteristics can support a formation that might otherwise cave into a well. This type of drilling fluid or mud plasters the walls of a well like mortar. Prevents entry of formation fluid into the well also prevent loss of drilling fluid into the porous zone. Formation of filter cake or mud cake.

5 Control subsurface pressure:

The pressures of gas, oil or water in formations penetrated by the bit may exceed the hydrostatic pressure of the fluid column in a well. If this happens, formation fluid will enter the well (kick). To kill a kick, the blow out preventer (BOP) are closed to hold back – pressure on the column of the hole to overcome the formation pressure. When formation pressures are expected to be high, a high mud weight is needed, so the pits and other equipment should be arranged to handle the heavy mud. A mud weight of 16 – 18 ppg is considered heavy.

Hydrostatic pressure (psi) = $(0.052 \times h)$

6 Cuttings suspension

The ability of the mud to hold cutting when drilling is temporarily suspended prevents cuttings from falling on the BHA and causing stuck pipe. The property of the mud that ensures this is the Gel strength or thixotropic property of the mud. Most time during drilling it is essential to stop circulation may be to check mud flow, or to do minor pump repair. When this happens the mud must become thickened so as to keep the cutting in suspension, otherwise they will fall on the BHA and jam the pipe. To free such pipe may be time and money consuming.

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The Mud Pump

The function of the mud pump is to circulate the drilling fluid from the tank, through the drill stem, to the bit (where hydraulic power is expended for jetting), back up the annulus and back to the tank. The mud pumps are either duplex or triplex.

Pressure Control System

Drilling mud is of invaluable assistance in the maintenance of subsurface pressure during the process of drilling. The hydrostatic pressure of the mud normally expressed as:

$P_h = 0.052 \rho_m h$

Where: P_h = hydrostatic pressure of mud, Psi

ρ_m = density of mud, ppg

h = height of mud column, helps to balance the formation pressure preventing any disastrous consequences (ft).

Blowout Preventers.

There are two types of BOP i.e. i) the annular preventer, ii) the ram preventer.

The annular preventer is used to seal the annular space between the drill pipe and wall of the wellbore.

The ram preventer consists of:

Pipe ram- closes the annular space around the drill pipe.

Shear ram – Use to cut the pipe in an emergency to seal the hole.

Blind ram – Use to close the open hole when drill pipe is out of hole.

Choke manifold ----this is used to control the flow of well influx out of the wellbore.

POWER SYSTEM

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PRIME MOVERS

To run all of the machinery required by a rotary drilling rig for hole making, power is necessary. Power in this case means energy for doing work. The basic source of rig power is an engine called a prime mover. It is usually an internal combustion engine, it is either compression ignition (diesel fuel) or spark ignition (using natural gas or liquefied gas as fuel). A prime mover converts the energy released by the combustion of a fuel into the energy of motion.

The outstanding advantage of diesel engine over gas or gasoline engine is its thermal efficiency – that is, they convert a higher percentage of fuel energy into useful power than other type of engine.

Efficient operation of diesel engines depends on

1. Sufficient clean air supply
2. Clean fuel of good quality
3. Correct timing sequence for ignition
4. Adequate amount of cooling fluids circulating through the engine to dissipate the heat generated by engine operation.
5. Good lubrication.

HOLE MAKING

The technique of drilling is the same whether in offshore, swamp or land drilling.

The only difference being in how to overcome the water depth or swamp depth as in swamp drilling.

When the drill vessel has been positioned, the temporary guide which is the first piece of equipment is however lowered to the ocean floor. This provides anchor for the four guide lines and foundation for the permanent guide structure. The base has saw tooth legs to penetrate the ocean floor.

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Foundation pile (conductor string) 20 – 36” OD usually about 100ft depending on ocean floor conditions is the first string of casing to be set into position. Guide frames are used on the guide lines to guide the bit from the vessel.

The hole for foundation pile is drilled with sea water and return fluid with cuttings

spilled over the sea floor. On completion, it is filled with gel water mud to prevent sloughing and fill. The bit is pulled out and the foundation pile installed. It is immediately followed by the permanent guide structure. The pile is then cemented in place with the head providing housing for further attachments. The conductor casing (20") is cemented in place with the wellhead equipment installed. This is followed by the installation of the blowout preventer stack. Subsequent casings and tubing's are run through the B.O.P. and landed in the well head. The tubing running from the vessel to the wellhead is called a riser but it is sometimes called conductor. The use of drilling mud helps to contain the well flow and the increased pressures imposed on the formation by gas in the well bore just as the land drilling. In this case, the hydrostatic pressure of the mud (Ph) is given by:

$$Ph = 0.052 \rho_m (d + h + e)$$

Where:

Ph = hydrostatic pressure of drilling fluid, Psi

ρ_m = density of drilling fluid, ppg (lb/gal)

d = depth of well below the ocean floor, ft

h = water depth, ft

e = elevation of the drilling fluid return overflow above the water surface,

It is usually about 60ft for submersibles and 25ft for ship shapes. Formation fracture pressure is given by

$$Pf = C (w.h + r.d).$$

where :

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Pf = fracture pressure of the formation psi

C = percent of theoretical overburden pressure

Where the fracture initiates (usually about 70%)

w = fluid head of sea water in psi/ft (0.445)

h = water dept, ft

r = weight of overburden rock psi/ft (1.0)

d = depth of the well below the ocean floor, ft

In theory, when $Pf = Ph$, the formation will fracture

$$0.052\rho_m (d + h + e) = C (w.h + r.d)$$

Swamp drilling is almost the same as offshore (sealed) drilling but in this case, the conductor pipe will have to be driven down or rammed down before drilling out the inner part (swamp). Further drilling can now continue as in other type of drilling.

PIPE CONNECTION/DISCONNECTION AND CHANGE OF DRILLING

BIT

The pipe for next connection is kept in the mouse hole while the drill string is disconnected at the Kelly. It is then swung over to the mouse hole for mousehole

connection and then joined back.

In replacing a drilling bit, the Kelly is disconnected and kept in the rat hole. The pipes are removed in succession until the last to remove the drilling bit which is then replaced. All these are achieved with the help of the derrick man.

Drilling bits may be changed due to:

- (i) Rate of penetration reduction
- (ii) History of records of bits of same type
- (iii) Worn out teeth
- (iv) Change in formation type being drilled

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DRILLING HOLE PROBLEMS

FORMATION DAMAGE: When the pressure of the drilling fluid column (hydrostatic pressure) becomes higher than the formation pressure or in an unconsolidated formation, the fluid may find its way into the reservoir (formation). This fluid invasion may lead to the reaction of drilling fluid with the formation fluid damaging the formation by reducing the permeability and porosity of the reservoir rock near the wellbore. This leads to impairment of formation permeability which reduces the flow of formation fluid to the wellbore and most time completely stop the flow. This phenomenon is referred to as formation damage, which is represented by skin effect. This damage when it occurs in the pay zone hinders or prevents the flow of oil or reservoir fluid into the well bore or production tubing. This phenomenon can occur during any operation performed on the well such as drilling, completion, workover e.t.c

It is therefore important that a balance be reached between the formation pressure and hydrostatic pressure.

LOST CIRCULATION

Lost circulation is defined as the loss of substantial quantity of whole mud to an encountered formation. This is shown by complete or partial loss of returns (returning annular mud flow)

Occasionally, the drilling fluid gets lost in the hole especially when the hydrostatic pressure is higher than the formation pressure and or the fluid loss programme is not well planned. This means that the circulated drilling fluid fails to return to the surface for recirculation.

This can result in a more serious hole problem such as a kick, lost of mud (money) and well bore stability.

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KICK

A kick occurs when formation fluid enters the well bore, that is to say, is the entrance of formation fluid into the well bore. It occurs when the formation pressure is higher than the hydrostatic pressure. It may happen when a gas zone (usually over

pressured) is encountered. If it is due to gas, it is normally circulated out. Drilling mud is primarily used to control the influx of formation fluid into the wellbore during drilling.

BLOW OUT

This happens when a kick has become uncontrollable. The pressure may increase to an extent that all drilling fluids will be forced out of the hole. It may lead to loss of rig and personnel. It is usually advisable that all means should be used for controlling a kick to avoid the hazardous consequences of a blow out. The primary cause of a blow out is still the formational pressure, higher than the hydrostatic pressure.

HEAVING SHALE PROBLEMS

Some areas are characterized by shale sections containing bentonite or other hydrated clays, which continually absorb water, swell and slough into the hole. Such beds are referred to as heaving shale. They can cause pipe sticking, excessive solids build up in the mud and the hole bridging are typical resultant problems.

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Fig.:3.8 Showing Heaving shale / hole collapse

SOLUTION

1. Changing mud system to inhibitive type e.g. lime, gyp which reduces tendency of the mud to hydrate water sensitive clays
2. Changing to oil emulsion mud
3. Increase circulation rate for rapid removal of particles

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WEEK 10 - 12

CHAPTER FOUR

WELL COMPLETION

This is the process of preparing a well for production. This includes the provision of necessary attachments for well for subsequent production of oil and gas. The process of well completion actually starts during the drilling process. The casings that are cemented in place during the drilling help greatly in well completion.

SETTING OF CASING

During drilling process casings are cemented in position as drilling progresses. Setting depth for any casing will depend on the stress it can withstand. Thus, a proper design procedure for casings is carried out before they are set. Casings may therefore vary in number and type depending on the formation. Fig. 4.1 below shows a particular setting configuration for a typical well.

Function of Casings:

1. Giving the well a permanent shape and size (precisely known diameter)
2. It serves as base for subsequent wellhead attachments.

3. Isolate pressure zones and prevent cross flow between formations.
4. Preventing contamination of freshwater sands.
5. To prevent cave-in or washout of the hole
6. To permit selective production of the pay zone

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CONDUCTOR

PIPE 36" / 300'

SURFACE CASING 20"

/ 2000'

INTERMEDIATE

CASING 13 3/8/ 5000'

PRODUCTION CASING

95/8"

LINER STRING 7"

FIG.4.1: A TYPICAL CASED WELL SHOWING CASING STRINGS
ASSEMBLY

FACTORS INFLUENCING WELL COMPLETION

The basic consideration for a well completion method is that it is such that guarantees maximum profit with minimum cost of completion. The factors considered are broadly grouped into two viz: mechanical considerations and reservoir considerations.

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MECHANICAL CONSIDERATIONS:

Mechanical consideration of the well is the basis for all well completion operations and subsequent workover. Therefore, well completion design must be mechanically fit to suit specific well completion, area and field considerations. The factors considered here include:

- (i) Maximization of profit
- (ii) Keeping the installation simple
- (iii) Overall reliability
- (iv) Operating condition and safety.

RESERVOIR CONSIDERATIONS:

This deals with the location of the fluid in the reservoir, the properties of the reservoir and how best to produce it. Some of the factors considered here include: the production rate, reservoir drive, secondary recovery and sand control.

WELL COMPLETION METHODS

Once a well has been drilled and the casings cemented in place, the next stage is the final completion and production. This is after the necessary tests have been completed e.g. coring to determine the formation properties, logging to determine

the productive zones and drill stem testing to determine the productive capacity of the well. Casing depth above productive interval is mostly set from well to well correlations. In most cases, once the production casing is set in position, a completion rig, a smaller rig may be brought in.

Methods of well completion are basically classified according to the method of completion and the completion geometry.

Classification based on the completion; namely open hole and perforated casing completions.

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COMPLETION BASED ON METHOD OF COMPLETION

A) OPENHOLE WELL COMPLETION

Open hole completions are also termed 'barefoot' completions.

In this type of completion the casing is set in place and cemented above the productive interval(s). Further drilling extends the well bore into and/or through the reservoir(s) which is left uncased, hence the term 'Open Hole'. This method has its own area of application depending on formation characteristics. Open hole is mostly applicable in carbonate reservoirs.

Advantages include:

- (i) Lower initial cost.
- (ii) Cement contamination and skin avoided because casing is not run in the producing zone.
- (iii) The well can be converted to a line completion, perforated completion or deepened as may be required.
- (iv) With gravel pack, provides excellent sand control.

Disadvantages include:

- (i) Excessive gas or water production difficult to control, i.e. Less control over pay section.
- (ii) Frequent cleanouts may be required.
- (iii) Selective fracturing or acidizing very difficult.
- (iv) Require more rig time on completion.

Fig.4.2 below shows a schematic diagram of an open hole well completion.

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Fig. 4.2:- Open hole Well Completion

B) PERFORATED CEMENTED CASING COMPLETIONS

In a perforated cemented casing completion, the hole is drilled through the target formation(s) and production casing is run and cemented in the hole. Like the perforated liner completion method, the casing and cement are perforated to reach the reservoir(s), allowing communication of the well fluids to the well bore.

Another term for a perforated casing completion is the 'set through' completion.

Perforated cemented casing completion methods are: the single zone completion and

the multiple zone completion.

Cased and perforated completions are the most common types of completions performed today since they offer selective pay zone perforation and also allow selective stimulation, which is more difficult in open hole or uncemented liner completions.

Perforated casing completion method can also be classified based on the mode of production, which can be tubingless and tubing production.

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Figure 4.3 : Schematics of perforated cemented casing completions.

Figure 4.4: Schematic of a perforated cemented liner

Generally, perforated casing completions generally have the following advantages and disadvantage.

ADVANTAGES

- (i) Excessive gas or water production easily controlled
- (ii) Can be selectively stimulated
- (iii) Logs and formation samples easily available for decision taking
- (iv) Easily deepened

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DISADVANTAGES

- (i) Costly
- (ii) Not adaptable to special drilling technique to avoid formation damage
- (iii) Log interpretation may present some problem in determining the accurate position of the formation sand.

CLASSIFICATION BY COMPLETION GEOMETRY

SINGLE ZONE COMPLETIONS

When a hydrocarbon reservoir can sustain flow due to its natural pressure, flow may be either, up the tubing string, the production casing, or both.

Flowing wells that are equipped with a single tubing string are usually completed with a packer. Single zone completions may involve the downhole commingling of production from several intervals within that zone.

At the design stage, optimum tubing size for maximum long term flow rate, future artificial lift needs, and future workover operations should be considered.

Flow Coupling

Selective

Landing

Circulating Sleeve

Packer

No go Nipple

Perforations

Pay zone

Pay zone

Fig.: 4.5 schematics of single zone completion

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Tubingless completions

Casing flow completions are a particularly low-cost method in marginal flow conditions such as low rate gas wells.

Wells may be produced without tubing but the following reasons clearly shows why tubing should be included in well completion.

- (i) Better flow efficiency
- (ii) Permit circulation of well fluids, corrosion inhibitors or paraffin solvents.
- (iii) Provide multiple flow paths for artificial lift system.
- (iv) Protect casing from corrosion, abrasion or pressure.
- (v) Provide means of monitoring bottom hole flowing pressure.
- (vi) Installation of SSSV

MULTIPLE ZONE COMPLETIONS

When completion is done to cover two or more zones so that all can be produced from the same well, it is known as multiple zone completion. Factors generally considered in the selection of multiple completions are:

- (i) High production rate
- (ii) Faster payout and
- (iii) Multi reservoir control requirements.

There are various ways of completing multiple zones using single or multiple strings of tubing. Some are:

- (i) Single string with alternate completion arrangement;
- (ii) Two parallel string, dual packer,
- (iii) Parallel dual with two alternate completion,
- (iv) Single string, single packer (both tubing and annular flow)
- (v) Single string dual packer with crossover choke, etc.

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Fig. 4.6: Schematics of multiple zone completions

Disadvantages of multiple zone completions

1. The requirements for well servicing/workover operations become more frequent
2. They are more complex and time consuming.
3. Very often the costs are higher than the cost of drilling a second well would have been.

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Flow Coupling

Selective Landing Nipple

Circulating Sleeve

Blast Joint

Pay zone Pay zone

Landing Nipple

Pay zone Pay zone

Fig. 4.7:-Single String with Alternate Completion Arrangement

Alternate Completions

This made possible the production of a particular interval at the presence of others one at a time. That is selective production, while other zones can be produced at a later date. There are various types of completion that falls under this category; some of them are single completion with alternate and dual completion with alternate.

Single string dual completion (SSD)

This is the most basic type dual completion where production of the lower zone is up through the tubing and production of the upper zone is up through the casing/tubing annulus. Fig. 4.6a

Twin string dual completion (TSD)

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In a twin string dual completion, separate flow from two zones (more if commingled) can be maintained through two tubing strings and two packers. Fig. 4.6 b

Multiple string completions

In this design, separate flow from three zones through the tubing can be achieved using three packers. Fig. 4.6 c

Concentric string completions

Concentric strings require less clearance than multiple string completions and can often achieve a higher overall flow capability. Such completions yield high total production per well and generally reduces costs. See Figure 4.6d.

Disadvantages of concentric string completions

- That there are many more potential leak paths than in a simple completion, including communication between perforated intervals behind the casing
- That wire line operations are more difficult and
- Packers can become stuck due to fall out from higher producing zones
- Completions are difficult to install and are usually restrictive to production, due to the small tubing sizes, to be economically attractive.
- Difficulty of future remedial workover of such wells prevents their widespread use.

COMPLETIONS EQUIPMENT

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Completion equipment comprises of the following:

Packer systems.

A packer is a device used to provide a seal between the tubing and the casing. In conjunction with a properly designed completion string, this seal directs the flow of reservoir fluids from the producing formation up through tubing to the surface. There are basically two types of packer, these are:

Retrievable packers

These are generally lowered into the well bore attached to and as an integral part of the production tubing string. As the name implies, retrievable packers can be recovered from the well, usually by applying pull to the tubing.

Permanent packers

These can be lowered into the well bore and set on an electric wireline or on tubing after which the wireline or tubing is released from the packer mechanically. When set, permanent packer can only be removed from the well by milling through the slips, thus releasing the grip on the casing.

SLIDING SIDE DOORS

A Sliding Side Door (SSD) or Sliding Sleeve is installed in the tubing during well completion to provide a communication path between the tubing and the annulus when it is opened.

It is used to:

- 1) Bring a well onto production after drilling or workover by unloading, (i.e. circulating the completion fluid in the tubing out with a lighter fluid)

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- 2) Kill a well prior to pulling the tubing during a workover operation (reverse of the previous) and allow selective zone production in a multiple zone well completion.

A SIDE POCKET MANDREL (SPM)

This is a receptacle for wireline installed devices which can be used for circulation purposes. An SPM is installed in the tubing string critically positioned for its intended application.

The dummy valve is essentially a plug which is installed in the SPM when circulation is not desired, e.g. when pressure testing or when idle.

WIRELINE LANDING NIPPLES

Landing nipples are tubing string components designed to accommodate the installation of various wireline retrievable flow control devices. The most common flow controls are plugs, standing valves, chokes, pressure and temperature gauges and storm chokes.

There are two types of wireline landing nipples, no-go and selective.

FLOW COUPLINGS

In high flow rate wells flow couplings are installed above (and sometimes below) flow control devices, including SSDs and SCSSSV landing nipples, in the completion string to protect against internal erosion caused by flow turbulence.

BLAST JOINTS

Blast joints are installed opposite perforations (non gravel packed) where external cutting or abrasive action occurs caused by produced well fluids or sand. In actual

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fact they perform the same task as flow couplings but protect against external erosion. Blast joints are usually available in 10, 15, and 20 ft lengths.

CHRISTMAS TREES

A christmas tree (often known as xmas tree) is an assembly of valves and fittings which forms the top of the completion. It is connected to the tubing hanger spool and directs the flow of fluids from the production tubing(s) into the production flowline(s).

All wellhead units have specifications that indicate working pressure and hydrostatic test pressure. The head of a flowing oil well is equipped with strong steel fittings designed for making the wellhead airtight and controlling well operation. The tubing is suspended from the wellhead fittings. The lower part of the wellhead fittings mounted directly on the tubing is called the tubing head while the upper part of the fittings mounted on the tubing head is called a Christmas tree

CHRISTMAS TREE VALVES

Lower master valve (LMV)

The LMV is a valve utilized on all christmas trees for contingency to the to shut off well flow in the event of a leaking UMV or the connection between them.

Upper master valve (UMV)

The UMV is used on moderate to high pressure wells as an emergency shut-in system. The valve can be actuated pneumatically or hydraulically. The UMV valve is a surface safety valve and is normally connected to an emergency shut-down (ESD) system.

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Flow wing valve (FWV)

The FWV permits the passage of well fluids to the choke. This valve can be operated manually or automatically (pneumatic or hydraulic) depending on whether is to be included in the surface safety system design.

Choke valve

The choke valve is used to restrict, control or regulate the flow of hydrocarbons to the production facilities. This valve can be operated manually or automatically and may be of a fixed (positive) or an adjustable type valve. It is the only valve on the christmas tree that is designed and used to control flow.

Kill wing valve

The Kill Wing Valve permits entry of kill fluid into the completion string and also for pressure equalization across tree valves e.g. during wire line operations or prior to the pulling or opening of a sub-surface safety valve. The kill wing valve is usually operated manually.

Swab valve

The Swab Valve permits vertical entry into the well for well servicing such as wire line (slick line and electric line), coiled tubing.

This valve (often referred to as the lubricator valve) is operated manually and is the uppermost valve on the Christmas tree.

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PERFORATORS AND PERFORATING

Perforating is one of the most important completion functions in cased holes. It is necessary that there should be adequate communication between the wellbore and all zones to help in evaluation to optimize production. A poor perforating job may result in low productivity of a well.

Causes of poor perforating job include:

- (1) Lack of understanding of the requirements for optimum perforating
- (2) Inadequate control of given clearance particularly through tubing guns and
- (3) Awarding of perforating jobs on price rather than quality.

The commonest types of perforators are the bullet perforators and jet perforators.

Bullet guns 3½ in OD or larger are applicable in formations with compressive strength less than 6,000psi. They can be fired selectively or simultaneously by design. Nozzle velocity of bullet guns is about 3,300ft/s.

In the jet perforating process, an electrically fired detonator starts a chain reaction which finally engulfs the main explosive. The high pressure generated causes the metal in charged liner to flow. Continued pressure build up on the liner causes a needle-like high speed jet of fire particles to spew from the cone at a speed of about 20,000ft/sec. This constitutes the perforating mechanism.

Other perforating methods include: Hydraulic perforators and mechanical cutters.

TUBING

This is a string of pipe with a small diameter through which the well fluids flow to the surface. The pipe is usually installed inside the casing. The diameter ranges from 3.5" to 1.9".

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Functions of Tubing:

- (i) Easy to remove if problems arise.
- (ii) Isolation of produced fluids from the casing and easier control of fluids.
- (iii) It facilitates circulation of heavy fluids into the wellbore to control the

well.

(iv) For better production efficiency.

(v) It allows for inclusion of safety devices and artificial life equipment in completion design.

WELLS AND OIL AND GAS GATHERING SYSTEMS

Oil well fluids are usually a mixture of oil, gas, water and sediments at elevated temperatures and pressures as they are produced. The oil is a mixture of many hydrocarbon compounds including gas compounds. It should be noted that oils from different reservoirs have different physical and chemical characteristics. There are also non-hydrocarbon compounds such as carbon dioxide (CO₂), hydrogen sulphide (H₂S) and other mechanical impurities of sand particles carried along with the fluid. Water which usually mix with the oil to form emulsion has to be removed. At the surface, separation facilities are usually designed to separate this mixture into separate streams of gas and oil and safely disposable water. The operations normally performed at a surface separation facilities include:

- (1) Separation of the oil and gas.
- (2) Transportation of the oil and gas from the wells to the storage tanks and compression station.
- (3) Removal of water and mechanical impurities.
- (4) Breaking up of the emulsion, desalination and stabilization of the oil (i.e. removal of light hydrocarbon from it).

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The system in which all these operations are performed is called an oil and gas gathering system. The surface production facilities have to be designed in such a way as to turn the mixture into separate streams or streams of clean dehydrated oil and gas and safely disposable water and solids. The oil and gas can then be sent to refinery for further processing.

FLOWSTATION

The first part of the separation facility system the fluid flows into is called a flowstation. This is where we have the initial separating equipment. Infact the first equipment, the oil encounters is the separator. Separator may be high pressure, intermediate pressure or low pressure separator. The fluid is accordingly directed depending on the pressure from the well, chemicals for the breakage of emulsion may be injected at the flow station before the fluid. It is at the separator that the first separation of the oil from water takes place. Flowstation may also contain a test separator which is used for a well on test. There are three types of separator, the vertical, horizontal and spherical separators, in offshore environment vertical separators are used. While on land where there is a high gas -oil-ratio, horizontal separators are used. Spherical separator is seldomly used but for condensate or wet gas fields. Many flow stations usually have the heater treaters which also help in the

breakage of the emulsion after passing through the separators. Gases from the separators and heaters are directed into the gas scrubbers to dry up the water in the gas by preventing water from entering the compressor cylinders, gas is then flared or then processed for sales. Flow stations are usually called production platform in offshore situation, equipments may not be the same in all flow stations or platforms. Flow stations also have tanks (small) which

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acts as an interim storage of the oil from which water is further drained before the crude oil is allowed to flow to terminal.

Fig. 5.1 A typical vertical separator with storage tank

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TERMINAL

Terminals are the last point the oil flow into for final separation and treatment before being pumped out for refining or sale. Final tests on the oil to determine its quality are also carried out here.

Generally, the oil from the various platforms and flow stations is received into the tank farm through a connection of pipelines. Terminals also contain heater treater and or separators for separation of the oil from water and gas. The oil after settling for sometime is drained and the bottom sediments is usually sent to the heater treater. There is also the skimmer pit here to remove the oil that must have escaped with the water. Terminals also contain booster pumps in most cases to pump the oil through pipelines to refineries for processing or tankers for sale. Terminals may also contain gas scrubber and in fact more operation than that of the flow stations are carried out here. Some tests carried out here include meter providing, tank gauging, fiscalization, etc. The flow process of oil from the well to its field terminals is shown in fig. 5.2.

HP Gas DEHYD-

Sweetening SALES

HP RATION

COMPRESSION

LP

HEADER/

FIELD SEPARATION TREATED OIL SALES

DISPOSAL

WATER

TREATING

Injection

Fig.5.2:- The Processing of Fluid from Wells

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OIL AND GAS PIPELINES

Oil and gas are gathered and handled at oil field by means of a system of steel pipelines laid over the territory. These pipelines could be from very small diameters of 80mm (3.24inches) to several hundreds in diameter. Oil and gas from wells enter the flow lines to the flow station and then the tank farm. When the necessary tests and processes are complete, it then enters the major pipeline for sale or otherwise. The diameters of these pipelines depend on anticipated rate of gas or oil flow. The depth of the pipelines depends on local climatic conditions but generally, they are protected from corrosion by a covering layer of bitumen before they are laid in the ground. Other means of protection include paint in cathodic protection, and use of chemical inhibitors for internal corrosion control. Where the pipelines cross rivers, railway tracks, hard surface roads, etc. The pipes are put inside protective casings made from pipes of greater diameter or reinforced concrete overcoat. Small rivers are often crossed by means of trestles and strength of pipelines welds is increased with the aid of special couplings. Gas pipelines are also provided with condense and water traps at depressions along the route in which the water and liquid hydrocarbon separating from the gas accumulate. The traps are periodically cleaned by blowing down through specially provided stand pipes. Oil field pipelines are usually fitted with gate and check valves of various types at some specific places along the line for easy disconnection during repair work or servicing.

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CHAPTER FIVE

TANK FARMS AND OIL MEASUREMENT

INTRODUCTION

Tank farms are usually found in the terminals as discussed under terminal. Generally, a tank farm consists of many tanks linked with pipelines from production platforms or flowstations and also pipelines terminating at the loading line for sales or refinery from further processing. Most of the oil measurements are carried out in the terminals (tank farm). Presently, there are two types of roofs used in the construction of tank farms. These are the FLOATING ROOF tanks and the FIXED ROOF TANKS. Tanks are often built in parallel where they are many with the pipelines linking them, constructed along the middle with branches to each tank.

FIXED ROOF TANKS

These types of tanks are constructed with the roofs fixed to the main body. The roof of the tanks do not move even when the oil is pumped out of it. Fixed roof tanks entrap the vapour of the oil in the space between the liquid and the oil.

FLOATING ROOF TANKS

This is designed in such a way that the roof moved, with the level of the liquid. The roof is not fixed or welded to the main body but has certain constructions

incorporated in the designed to enable the roof slide along the mines surface of the tank as the level of the fluid fluctuates due to inflow or outflow of fluid.

All tanks where fixed roof or floating roof are calibrated either in feet or meters and their subdivisions. They have vents on the roof usually used for measurement.

They also have pipe linkages for pumping in crude and dealings purposes. Tanks may be of very large capacities ranging from hundreds to thousand of barrels and

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may also be several feet high. All have staircase for climbing them for any operation to be carried out.

OIL MEASUREMENTS

The measurement done to ascertain the quantity and quality of crude oil in a tanks are generally classified under a process called fiscalization. This includes gauging (i.e. tank dipping) sampling, lab analysis and calculation to determine the quantity.

TANK DIPPING

This is a method of ascertaining the level of oil in a tank using approved steel tape. The tape usually has a plumb bob mounted on a swivel. This keeps the measurement steel tape released always in tension. The graduation starts from the plumb bob (a graduated cylindrical piece of metal). Tanks are usually allowed to settle after pumping into it before dipping (at least 3 hours). The tape is lowered into the tank and the level of the liquid out recorded as it touches the datum plate. This is repeated until two consecutive identical readings are obtained.

SAMPLING

This refers to the method of obtaining representative sample of crude oil for quality determination. The sampling apparatus is called the “thief can”. The sample is usually withdrawn with the oil in the thief can and put into a sampling can.

Sampling is usually conducted immediately after dipping. The number of samples taken from a given tank depends on the height of the crude oil in the tank.

(10ft) or below - One sample

(10 - 15ft) - Two samples

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5m (15ft) or above - Three samples

After sampling, the temperature of the crude oil is usually taken and the average tank temperature recorded before the sampling is poured into the sampling can. Before sampling is sent to the lab for analysis, it is labeled as follows:

- (1) Tank number
- (2) Name of sample: light, heavy medium
- (3) Upper/middle/lower/composite
- (4) Temperature of tank

LABORATORY ANALYSIS

The following parameters are usually determined in the laboratory. These are:- The

basic sediments and water (B.S & W); A.P.I. gravity and temperature at which the gravity was taken.

DETERMINATION OF A. P. I. GRAVITY

The hydrometer has a thermometer combined with it. The hydrometer measures or determines the API gravity while the thermometer determines the temperature at which the A.P.I. gravity was measured. To determine this, some quantity of the crude oil is poured into a 500ml cylinder after agitation in the sampling can and the top cleared off foam if any. The combined hydrometer – thermometer is then inserted into the cylinder and the A.P.I. gravity is then read when the system reaches equilibrium. It is removed and the temperature read from the thermometer graduation. It is recorded as A.P.I. at XOF.

141.5

□□□ □ = – 131.5

□. □@600 □

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DETERMINATION OF B. S. & W.

This experiment helps to determine the percentage of water and other impurities in the crude oil. This is usually subtracted to get the net volume of oil in a tank or net volume loaded by a tanker.

The commonest method for the determination of B. S. & W. is the centrifuge method. In this method, 50ml of the sample is put into a cone shaped graduated centrifuge tubes of 203mm. 50ml of toluene or xylene is added as a reagent. It is usually saturated with water. After each tube has been corked, it is well shaken and put into a water bath up to about 100ml mark. The temperature is raised up to 120°F and left for 10mins. The centrifuge is then switched on. The tubes left for 10mins and after this it is removed. The B. S. & W. is read off from the segregation at the bottom to the nearest 0.05 or 0.1 depending on whether it is lower than 1 or greater than 1.

CALCULATION OF QUANTITY OF CRUDE OIL IN A TANK

Each tank has its own calibration table. This is used with ASTM tables. This is used in conjunction with ASTM tables and the results got from gauging and lab analysis to calculate the exact quantity of crude oil in a tank.

The tank dip is referred to the tank calibration table to get the gross barrels at the tank temperature. The A.P.I. at the tank temperature is corrected to A.P.I. at 60°F using tables also. The volume correction factor, the long ton per barrel factor is got from the tables using the A.P.I. gravity as a guide. These factors are subsequently used to get the U.S. Barrels and long ton barrels. In floating roof tanks, the roof displacements can be calculated using the roof displacement tables for each tank. The net water is also calculated and subtracted from the gross barrels to get the net barrels.

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Although procedure may vary from place to place. ASTM tables are used in all cases and the above description represents this country's system of calculation. All calculations are done on a neat sheet or form.

CALCULATION OF DAILY AND MONTHLY STOCK

The procedure for the above calculation is carried out every morning usually at 6.00 hours, to ascertain the quantity of crude oil in a given tank. This will then show how much crude oil come into the tank farm every morning and how much also leave for exportation or processing. This morning calculation is called daily morning stock taking.

At the end of the month, the summary of the daily stock taking is clearly written out. This is called monthly morning stock. This is in form of a balance sheet as it summarizes the total inflow of all crude oil into the tanks and the total flow for a particular month.

In some cases, the calculations above may be closely monitored by metering in a unit called the LACT unit (Lease Automatic Custody Transfer Unit) which registers the oil output from tanks or input into the tanks.

SEALOADING LINES AND FLOATING BUOY MOORINGS

Crude oil is pumped from the tanks after fiscalization to the main export lines in case of exportation. In offshore situations as it is in most cases, they are called sea loading lines. Usually these lines terminate at the Buoy Mooring (SBM) via the Berth Operations platform. The single buoy mooring is a device which facilitates offshore loading of big tankers which cannot come near to land because of the depth of water required for it to do so. The tanker is hooked to the single buoy mooring while being loaded from it.

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THE LACT UNIT

The measurement of oil on transit is now being automated through the lease automatic custody transfer unit. It is a unit or system for measuring and transferring the oil from the gathering system into the pipeline. This unit registers the amount of quantity and quality of the oil being transferred automatically as the oil is being transferred.

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